How People-Pleasing Affects Social Learning in Friendships

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Abstract

Maintaining close friendships depends on learning each other's preferences through emotional feedback. However, in the case of people-pleasers, they often express positive emotions even when they feel negative emotions. This introduces systematic bias into the signals that others perceive and rely on for social learning. In this work, We model this bias using Bayesian learner agents that interacts with each other under different levels of emotional masking. We use both controlled toy simulations and naturalistic emotional dynamics from the MELD Friends dataset, we then try to determine how people-pleasing change and affect what others learn. In these simulations, we identified that masking reduced the informational quality of the expressed emotions, it also produced high confident agents despite reduced accuracy in the predictions, and finally generated mental models that failed to generalize across communicative styles. These results suggest that even mild people-pleasing can create misunderstandings by giving learners the illusion of accurate understanding while quietly degrading the true structure of social knowledge.

Keywords: social learning; Bayesian inference; emotion masking; people-pleasing; affective dynamics; communication bias

Introduction

To maintain a friendship, both parties need to learn each other's preferences to keep each other happy. This learning process happens through events when one chooses a restaurant, an activity, etc. Then, we rely on feedback to infer future decisions. Yet in the case of a people-pleaser we may note that their feedback is distorted: when a people-pleaser displays positive emotions that contradict their true feelings they introduce an information problem. And if every time someone says "yes, that was amazing" they are telling us the same thing, we will never get to know the true preference of a person. Over time, that feedback loop drives both people farther from reality: the people-pleaser becomes unhappy; their friend gains confidence from a false model. This drives the research hypothesis to three questions: Why do misunderstandings happen even when everyone is trying to be nice? Does being "too positive" make us harder to read? Can two people believe they understand each other when their signals are quietly misaligned?

Existing research in social cognition argues that people build accurate mental models of others' emotional dynamics through Bayesian learning. For instance, if someone feels joy, then it is more likely for them to feel calm than angry. Thus, people leverage knowledge about these regularities to form predictions (Thornton & Tamir, 2017; van den Bos et al., 2022). These models enable prediction of friends' future preferences and emotions, and prediction accuracy correlates with relationship quality (Zhao et al., 2022). However, these models, or fairly all Bayesian models of social learning, assume that observed behaviors provide true informative signals about internal states. For example, if a person smiles, the model assumes it reflects their true internal state of happiness. But in people-pleasing, this is not the case. A people-pleaser is defined as a person who feels a strong urge to please others, even at their own expense. Their behaviors are systematically biased, often masking negative feedback with positive signals. When feedback is systematically biased in this way, the fundamental assumption of informative observation breaks down. While research has examined noisy feedback in other learning contexts (Im & Zhang, 2024), no prior work has modeled how this bias accumulates to degrade preference learning in friendships.

We address these questions using emotional dynamics from the MELD dataset, which contains emotion-labeled dialogues from the TV show Friends. First, we operationalize people-pleasing as "masking rate"—how often a character responds positively after negative utterances. Then, we extract character-specific emotion transition matrices that represent how each character's emotions naturally evolve over conversations. Using these real patterns, we build Bayesian learner agents that observe expressed emotions and try to learn the underlying dynamics. In our simulations, we measure three things: (1) accuracy of learned models compared to true dynamics, (2) confidence in predictions, and (3) similarity between what both agents learned. This lets us isolate how biased emotional signals affect social learning using real human communication patterns.

All together, these presented challenges refer to a much broader question: how does biased emotional feedback reshape the mental models we form of one another? In this work, we acknowledge that humans, by nature, sometimes give biased emotional feedback. However, we will focus on people-pleasing as a case study, as they are known to mask

their true feelings much more than the average person. To investigate this, we use real conversational data with controlled simulations to determine how people-pleasing changes the signals available for social learning. Our approach allows us to isolate the core mechanisms through toy simulations and then test whether the same patterns appear in naturalistic emotional dynamics. In the Results section below, we show that this framework reveals three main insights: (1) people-pleasing behavior reduces the quality of emotional informational signals. (2) It creates a divergence between confidence and accuracy, so-called (Confidence Trap), where the person is much more confident about inaccurate information or a prediction. (3) It produces mental models that fail to generalize across different communicative styles, such as mental models for (pleaser-to-pleaser versus pleaser-to-honest)..

Background

Social Prediction & Mental Models:

Current work in cognitive science argues that people are able to predict others' emotions and actions. Research on action prediction by (Thornton & Tamir, 2017) highlights that people can accurately estimate the transition probabilities of everyday behaviors. In simple terms, a person's current action is often a useful clue to what they will do next: angry people aggress, tired people rest, and so on. Across five preregistered studies using surveys, and other methods, the researchers found that people's predictions closely matched real action-transition statistics, with correlations ranging from P-value = 0.34 to 0.38 at the item level. They argue that people draw on structured conceptual knowledge: actions that sit closer together in a conceptual space are assumed to be more likely to follow one another. The main takeaway is that human knowledge seems to be organized around prediction. The mind builds internal maps of how certain states change over time, allowing people to make reasonable guesses about what will happen next based solely on the current state—whether those states are actions, emotions, or social behaviors.

Building on this work, (Tamir & Thornton, 2018) shows that the approach that people follow when inferring predictions on others' mental states is based on an internal structure. This internal structure as they call it the "3D Mind Model" which is rationality, social impact, and valence. They defined rationality as whether a state is more cognitive or more emotional. Social impact as whether a state is high intensity and socially directed or not such that the more likely someone is to engage with you (sociality) and the more intense their mental state (arousal) the more likely they are to have a substantial impact upon you. Valence indicates the extent to which a state is positive or negative when we experience it. Thus, we can conclude that emotions experienced or perceived are not random but sequential, they fit within a psychological space that treats similar states such as angerness and frustration as closely related states and calmness and contentment as states that are far away from the following states.

Another paper by (Zhao, et al., 2022) briefly supports the ideas proposed by Tamir, examining how different people can predict their friend's emotions and how this relates to the quality of their relationship. Based on that, they are testing if people who know each other have more accurate internal models of the other friends' patterns when expressing their emotions over time. They found that friends who knew each other longer were significantly better at predicting each other's emotional reactions through repeated interactions.

Bayesian Learning in Social Contexts:

One research paper led by (Griffiths, et al. 2006) hypothesizes that Bayesian inference is the core idea behind how people predict events in everyday life. In simple terms, a person combines what they already know about events with new evidence to make smart guesses about the world. The researchers conducted a study where they asked participants to make predictions based on partial information they were given. The predictions were in the form of questions such as how long a person will live, how much money a movie will gross, etc. They then compared participants answers with an optimal Bayesian model that used real statistical information about these events. The predictions of the participants matched the model s outputs closely, which implies that people rely on prior knowledge in addition to the observed data they were given when making predictions; However, another paper led by (Li et al., 2024) explored how biased feedback affects learning in recommender systems. These recommender systems are designed to generate meaningful recommendations for users based on items or products that might interest them. The researchers noticed that the ratings in their data have many inconsistencies between the observed ratings and the true preferences; they called this outcome measurement error. As a result, the observations used by the learner do not represent the true underlying preferences accurately, and the model updates its beliefs based on misleading information and forms distorted models. From this paper, we conclude that the success of the recommender model, or any learner, depends on the information it receives, and when those signals differ from the underlying state, the inference process becomes biased. Connecting this paper with the previous section, we can assume that this also happens with humans in the context of emotional learning in social relationships. However, in this case, the biased data comes from the people-pleaser s feedback, where they do not represent their underlying state especially if it is a negative internal state and replace it with positive expressions consistently, creating inaccurate belief updating for the people around them. However the final question would be: how can people make reliable predictions in such a complex space?

One paper that relates to this question, led by (Bertolero and Griffiths, 2014) from course sample papers, studies whether prediction and learning are possible in high-complexity environments. They emphasized that social and emotional states often affect or influence each other. The authors also ad-

dressed the broader concern of having this holistic structure that makes it hard for the learner to form accurate models. The researchers showed that humans can make reliable inferences in complex, interconnected environments as long as these environments have some regular underlying structure within them. In simple terms, the researchers believe that the complexity of social or mental spaces doesn't prevent learning. They believe that the learner succeeds by identifying the patterns that organize this space. This analysis supports the idea that Bayesian learners are also capable of capturing the pattern and modeling emotional understanding even if the emotional model is complicated. This suggests that all that matters is whether the complex structure offers consistent patterns for learners to pick up on.

People-Pleasing as Biased Feedback:

One research study by (Gross, 1998) on emotion regulation shows that human beings do not always express what they feel. He described what is called "expressive suppression," which is defined as minimizing an outward expression while experiencing this emotion internally. The suppression of emotions reduces the clarity of emotional signals, even if it helps maintain harmony in social spaces. It is linked to poorer interpersonal outcomes and reduced mutual understanding. Recent descriptions of people-pleasers extend this logic to specific interpersonal patterns. In this research question, we address people pleasing as a one way of studying how bias in learning models happen. However, people pleasing is the extreme behavior of emotional masking, which makes it interesting to study as they are overly positive and care about pleasing others in a very excessive way that we can capture from the data. More details on people pleasing are below:

In an article written by Marissa Moore (2024), she describes people-pleasers as individuals who disregard their own needs to please others, have difficulty saying no, or say yes to things they don't want to do, often go to great lengths to be liked, avoid disagreements, and mitigate the feeling that they will be abandoned. When people-pleasers experience hurt or disappointment, they frequently avoid this conflict and express agreement and reassurance instead. Thus, from a learning aspect, people who interact or are friends with people-pleasers accumulate inaccuracies in how they understand people-pleasers' emotions and predict their reactions.

Approach

The goal of the approach that we will be following in the paper is to primarily represent or formalize people-pleasing behavior as a distortion of emotional signals using a Bayesian social learner: How does this kind of biased feedback affect what others learn?

To make things easier to comprehend, let's assume that person "P" is a people-pleaser who exhibits the primary characteristics of this behavior. In this model, we will separate the emotional signals that people observe about P from his internal emotional state. We will assume that each agent (including P and other people) has an internal emotional state that evolves over time according to a fixed transition structure, which can be represented as a transition matrix over a small set of discrete emotions. The transition matrix is defined as the probability of a person transitioning from one state to another (for example, probability of being happy after being sad). The transition matrix is expected to represent the true underlying emotional pattern of the person, regardless of whether they choose to display it or not. How we computed these dynamics using transition matrix from data will be later explained in the methods section.

We define learner agents as brains that try to understand and learn from a person's emotional patterns from their expressed emotions (what they observe). Each agent learner observes a sequence of emotional signals expressed by their interaction partner over time. The learner tries to build a mental model that observes how this person's emotional states change and their associated probabilities. The agents' learning rules are relatively simple and Bayesian-like. After learning, we try to quantify the masking rates for the agents which is the probability they will mask their negative emotions with positive ones. We study the results and consequences of masking in a simulation where two agents interact with each other. In these simulations, each agent exhibits a sequence of internal emotional signals, passes them through masking, and then expresses the resulting emotions. At the same time, each learning agent builds a mental model of the other agent based on the feelings expressed by that agent. After many iterations of learning, we compare the mental models built by learning with the actual mental models of those agents. We focus on three important points: how accurate the mental models are, how confident the learners are in their predictions, and how aligned the two predicted mental models are.

In addition to simulations grounded in real conversational data, we also run simplified toy simulations to isolate the model's core mechanisms. These toy simulations use abstract emotional worlds with controlled transition structures to highlight three effects. [First, there is a reduction in signal quality when people-pleasing is present. Second, a "confidence trap" occurs in which subjective certainty remains high even as learning becomes less accurate. Lastly, there are failures of generalization between honest and people-pleasing communication styles.] Together, the dyadic and toy simulations provide complementary views on how systematic emotional masking can create an illusion of understanding while degrading the actual accuracy of social learning.

Method

We used the MELD dataset to apply the approach, which contains about 1,433 dialogues from Friends TV show. The dataset contains dialogues and emotion labels for each utterance. Here, we will focus on the six main characters (Ross, Rachel, Monica, etc.) as their dialogues and utterances are statistically significant enough to infer results. Based on the data, we are working on seven main categories of emotions: joy, sadness, anger, surprise, fear, disgust, and neutral. we

then assigned valence scores for the different categories of positive and negative emotions: [joy = 1, surprise = 0.4, neutral = 0, sadness = -1, anger = -0.8, fear = -0.7, disgust = -0.6]. These values followed existing research on affective science (Russell, 1980; Warriner et al., 2013), in which positive emotions were assigned positive values, neutral emotions were assigned 0, and negative emotions were assigned negative values. For each Friends character, we computed a transition matrix, which we defined before, but to recap, it is the probability they transition from one emotion to the other:

$$P(\text{emotion}_t \mid \text{emotion}_{t-1})$$

We then tracked the transitions of emotions from state i to state j for each character in the show across all dialogues and counted how many times each transition occurred for each person. The matrix was computed as:

$$T_{\text{true}}[i,j] = \frac{N_{ij} + \alpha}{\sum_k N_{ik} + 7\alpha}$$

Where N_{ij} is the number of times a person transitioned from state i to j, and α is the Laplace smoothing parameter to handle transitions that did not happen in the data. The resulting matrix is 7×7, representing true emotional states before masking.

After, we calculated the masking rate, where we counted the times each character responded with positive emotions after someone expressed a negative emotion:

$$r_{\text{baseline}} = \frac{\text{count(positive response} \mid \text{negative context)}}{\text{count(all responses to negative contexts)}}$$

After calculating the initial masking rate, we then added an environmental factor that represents different levels of social pressure ($\lambda \in 0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5$) where low λ reflects a safe environment and higher values represents social pressure that forces people to mask their feelings more. Thus, in each step: if the agent has a negative state, then the probability they will mask it is:

$$P(\text{mask} \mid s_{\text{neg}}) = \min(r_{\text{baseline}} \times \lambda, 1.0)$$

To test how learning differs based on masking rates and transitions, we created a simulation with two agents (A and B) where they learn and interact with each other. Each agent is supplied with three main things: their own transition matrix T_{true}, their masking rate, and a learned transition matrix of the other agent which are initially initialized with a Dirichlet prior (all 1s). At t = 0, each agent starts with an emotional state randomly sampled from the stationary distribution of T_{true}. Then the simulation is run for 240 time steps because the initial results showed that confidence is converged after 150 steps and accuracy after 200 steps.

At each step of the simulation, 4 main things happen: (1) The agent chooses their new mental state based on the probability of the next emotion based on the true matrix. For example, if the agent's initial state was sad, we count the probability of the next emotional state: is it more likely to be sad/angry/neutral? (2) If the current mental state of the agent is negative, we then calculate the probability they will mask this feeling with: P(mask) = $r_{baseline} * \lambda$ where $r_{baseline}$ is the normal masking for the agent, and λ is the environmental stress. Then based on the P(mask), the agent decide their decision to mask or not based on a Bernoulli draw (3) After that, each agent updates their learning model on the other agent, and if masking happened, it means they did not learn accurately. (4) Finally, on each simulation step, agents update their emotional states according to their transition probabilities. Simultaneously, each agent updates their learned model of the other agent's emotional dynamics by incrementing a counter for observed transitions:

$$\text{count}[s^{\text{obs}}_{\text{prev}}, s^{\text{obs}}_{\text{current}}] \mathrel{+}= 1$$

The learned matrix was then normalized row-wise:

$$T_{\text{learned}}[i,:] = \frac{\text{count}[i,:]}{\sum_j \text{count}[i,j]}$$

For example, while Agent A updates their emotional state, Agent B constructs a learned transition matrix for Agent A.

In addition to character-based simulations, we then built a Toy Simulation to isolate complexities from the dataset and build an artificial world: we build a 7*7 transition matrix with minimal elevation in diagonal values of 0.2; and elevated the value of emotional inertia such that if an agent is happy, probabilities are higher that his next stage he will remain happy. And for the other probabilities, it was equally distributed across same valence transitions. We then defined the two main agents: Honest agent with masking rate of 0 and a pleaser agent with masking rate of 0.3. We tested four training scenarios: Honest-Honest, Pleaser-Pleaser, Honest-Pleaser, and finally Pleaser-Honest where the first agent learns the preferences of the second with 240 time steps, and ran additional 100 steps independently to confirm. We then measured accuracy (proximity to true dynamics), confidence (subjective certainty) , similarity((alignment between agents' learned models)), emotion entropy, and valence variability with the following formulas:

$$\text{Accuracy} = 1 - \frac{\|T_{\text{learned}} - T_{\text{true}}\|_F}{\|T_{\text{true}}\|_F}$$

$$\text{Confidence} = \frac{1}{7}\sum_i \max_j T_{\text{learned}}[i,j]$$

$$\text{Similarity} = \frac{\text{vec}(T_A) \cdot \text{vec}(T_B)}{\|T_A\|_F \times \|T_B\|_F}$$

We also measured signal quality from expressed sequences: Emotion entropy $= -\sum_i p(e_i) \log p(e_i)$, and Valence variability = SD(v_t).

Results

Here, we tested two simulations: Toy and character-based with two main agents over different conditions, which will be highlighted in three main findings below. To elaborate, the toy model is merely a test model with predefined assumptions, as explained in the previous sections (isolating the core mechanism with larger effects), and the character-based model tests these assumptions on real-world data to see whether these patterns exist.

Finding 1: Signal Quality Degradation

The first finding is quantifying how people-pleasing affects the signals of expressed emotional states. In the controlled toy simulation, illustrated in Figure 1 below, we found that pleasers reduced signal quality by 45% in valence variance, compared to the honest agent (Honest: 0.32 vs. Pleaser: 0.17), with $p < .001$. And 20% lower emotional entropy (Honest: 3.0 bits, Pleaser: 2.4 bits) with p ¡.001. Thus, we can conclude from the findings that people-pleasing smooths emotional signals by masking negative states with positive expressions, thereby affecting learning.

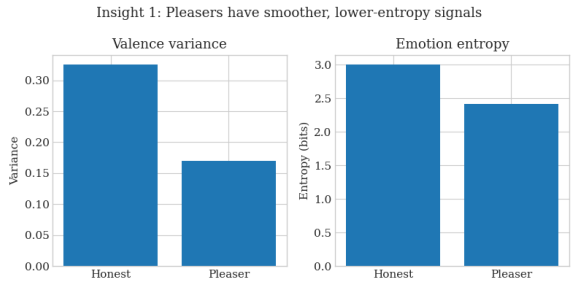

Figure 1: Signal quality comparison H-P.

The Confidence-Accuracy Divergence

We then quantified and examined how the masking by the pleaser affects learning. In the toy simulation, as you can see in Figure 2, the pleaser showed 46% higher confidence than honest agents, specifically: (Pleaser: 0.39, Honest: 0.27) with $p < .001$. This means that the people pleasers are more confident about understanding their friends' emotions but with less accuracy which creates a confidence trap.

Figure 2: pleasers vs honest (confidence & accuracy)

This confidence vs. accuracy problem also appeared in the Friends character-based simulation: Here, we examined two main pairs: (Joey & Phoebe who have close masking rates) and (Monica & Chandler - contrasting masking rates). From Figure 3 and 4 below, we can infer the following:

1. Joey & Phoebe Case: the accuracy has declined from 99.50% to 99.15% as environmental masking stress increased from 0 to 1.5, which introduces the same pattern with less magnitude as our toy model.
2. Monica & Chandler Case: it also showed similar trends with 99.65% to 99.20% .

So in conclusion, the controlled simulations demonstrated large effects. However, the character-based simulations illustrated that these patterns exist in the natural dynamics.

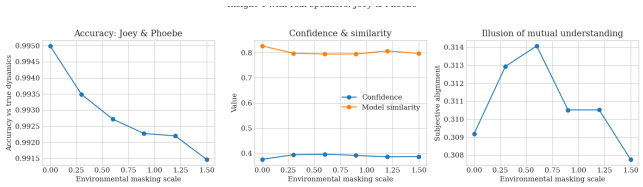

Figure 3: Joey & Phoebe - Friends validation

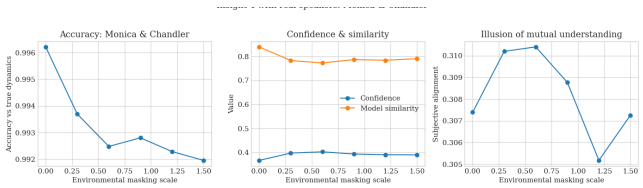

Figure 4: Monica & Chandler - Friends validation

Transfer Between Honest and Pleaser Worlds

Finally, we tested if models that learned in one style such as only an honest or only a pleaser learning would also generalize to the other? The results below in Figure 5 toy's model provided the following insights:

Case 1: when an honest person learns from a similar honest person, the performance of learning is high.

Case 2: Similarly, in the case of a pleaser learns from a pleaser, the performance of learning is high.

Case 3: When an honest person learns from a pleaser, the accuracy drops.

Case 4: When a pleaser leans from an honest, the accuracy drops even more.

This concludes that the learning model for pleasers cannot be generalized or adapted in the honest learning model and vice versa. Thus, we can see from the following that, pleasing does not only drop the signals and distort the predictions, but it also only works well with other fellow pleasers in the same learning model.

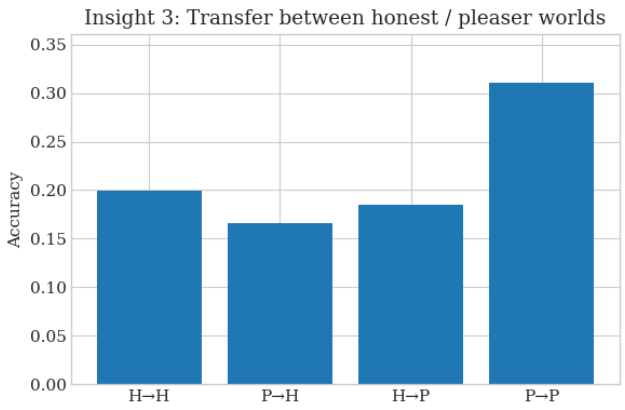

Figure 5: Transfer between Honest vs. Pleaser World

Discussion

This study began with a simple question: Does continuous people-pleasing (the expression of positive emotions rather than negative ones with others) affect how people learn about each other's emotions? Among both the toy simulations and the Friends data, we found converging evidence that emotional masking biases what others learn.

First, in the toy simulations, we clearly isolated the core mechanisms: masking reduced signal quality. In addition, it inflated confidence and led to specific learning strategies that only work within the same agent's learning types, such as honest-to-honest versus pleaser-to-pleaser. However, these significant effects are expected in such a controlled environment where masking is the primary driver of that distortion.

On the contrary, the Friends-based simulation illustrated the same qualitative patterns, but with much smaller magnitudes. This attenuation could be explained by the fact that the extracted emotional behavior from the MELD dataset was noisy. The masking rates were low and varied modestly across the characters, and many additional factors influenced expression. However, we observed that accuracy consistently declined as the masking pressure increased, while confidence remained nearly unchanged, mirroring the confidence-accuracy trap described in the toy model.

Together, the presented results suggest that even mild people-pleasing can undermine social learning process as it smooths emotional signals, produces overconfident and inaccurate mental models, and also creates representations that do not transfer well across different world contexts (pleasing vs. pleasing and pleasings vs. honest).

In the end, this study has several limitations. First, the emotional data comes from scripted dialogue rather than naturalistic conversation; the designed model abstracts away much of the richness of human affect; and masking was calculated using a simple behavioral heuristic. It should be considered more, since the world is much more complex; however, this does not undermine its importance. Future work will examine larger social networks. It will also incorporate richer linguistic cues to obtain a more accurate representation of the masking rate and to test how learners recover once masking decreases. Overall, the findings highlight a subtle but consistent effect: being "too positive" can create the illusion of mutual understanding, even as the accuracy of that understanding deteriorates.

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References

Thornton, M. A., & Tamir, D. I. (2017). *Mental models accurately predict emotion transitions*. Proceedings of the National Academy of Sciences, 114(23), 5982–5987. Available at: https://www.pnas.org/doi/abs/10.1073/pnas.1616056114

Tamir, D. I., & Thornton, M. A. (2018). *Modeling the Predictive Social Mind*. Trends in Cognitive Sciences, 22(3), 201–212. Available at: https://www.sciencedirect.com/science/article/pii/S136466131730267X

Gross, J. J. (1998). *The Emerging Field of Emotion Regulation: An Integrative Review*. Review of General Psychology, 2(3), 271–299. Available at: https://doi.org/10.1037/1089-2680.2.3.271

Zhao, Z., Thornton, M. A., & Tamir, D. I. (2022). *Accurate emotion prediction in dyads and groups and its potential social benefits*. Emotion, 22(5), 1030–1043. Available at: https://doi.org/10.1037/emo0000890

Griffiths, T. L., & Tenenbaum, J. B. (2006). *Optimal Predictions in Everyday Cognition*. Psychological Science, 17(9), 767–773. Available at: https://doi.org/10.1111/j.1467-9280.2006.01780.x

Li, H., Zheng, C., Wang, W., Wang, H., Feng, F., & Zhou, X.-H. (2024). *Debiased Recommendation with Noisy Feedback*. Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), 1576–1586. Available at: https://doi.org/10.1145/3637528.3671915

Moore, M. (2024). *People-pleasing: What it is and how to stop*. Medical News Today. Available at: https://www.medicalnewstoday.com/articles/people-pleaser

Russell, J. A. (1980). *A circumplex model of affect*. Journal of Personality and Social Psychology, 39(6), 1161–1178. Available at: https://doi.org/10.1037/h0077714

Warriner, A. B., Kuperman, V., & Brysbaert, M. (2013). *Norms of valence, arousal, and dominance for 13,915 English lemmas*. Behavior Research Methods, 45(4), 1191–1207. Available at: https://doi.org/10.3758/s13428-012-0314-x